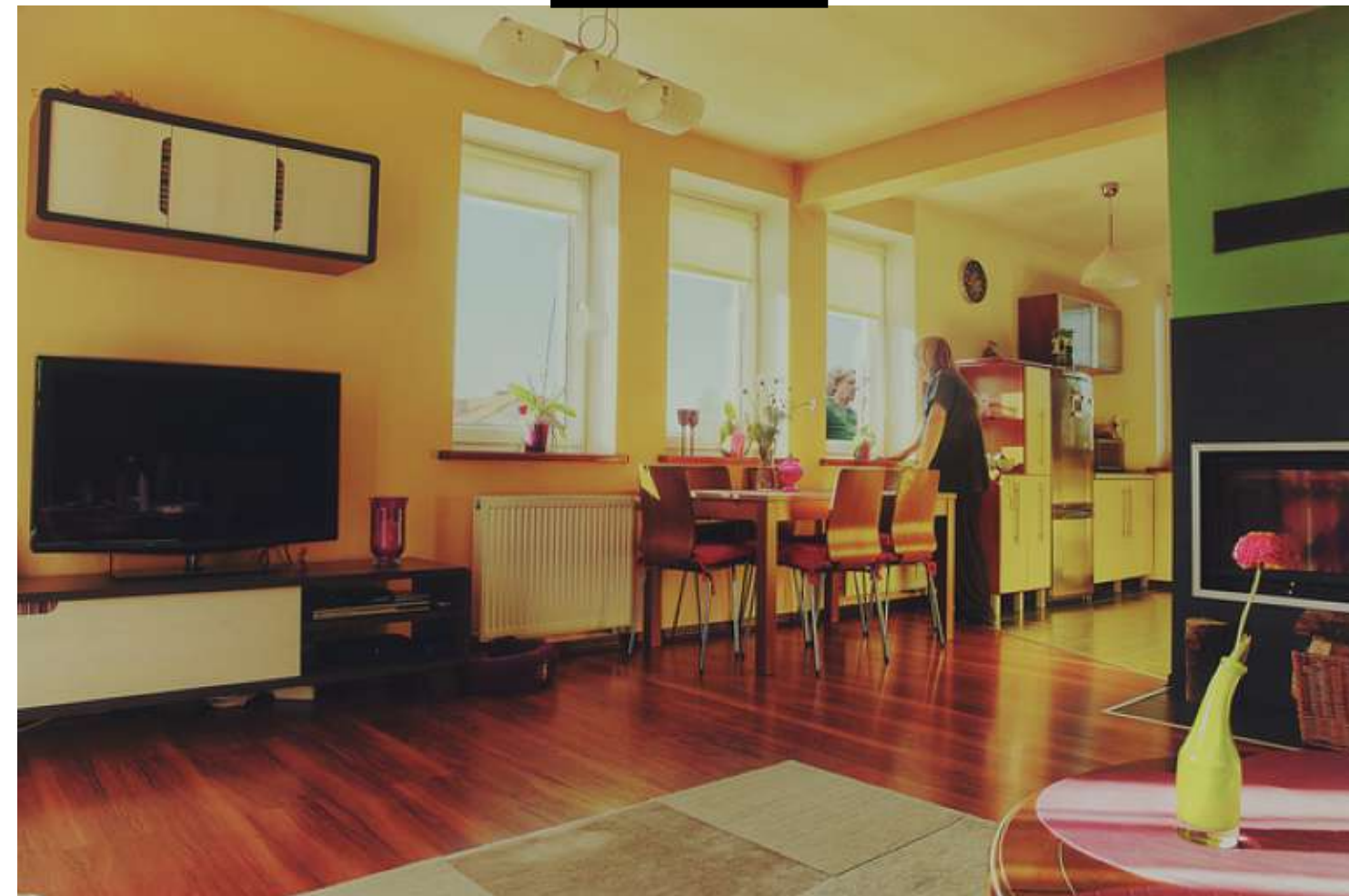


Figure 4.2: CRATTT vs Baseline — Category Summary
N=20 images | $\tau=0.321$ | $\alpha=0.4$ | $\beta=0.6$

Category	Baseline mAP	CRATTT mAP	mAP Delta	Mean Rejected
Blur	0.226	0.182	-0.044	3.719
Digital	0.382	0.291	-0.091	4.075
Noise	0.324	0.259	-0.065	3.725
Weather	0.461	0.377	-0.085	4.381

Corruption: Gaussian Noise | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000139.jpg



BASELINE (GroundingDINO)
Detections: 8 | mAP: 0.0744

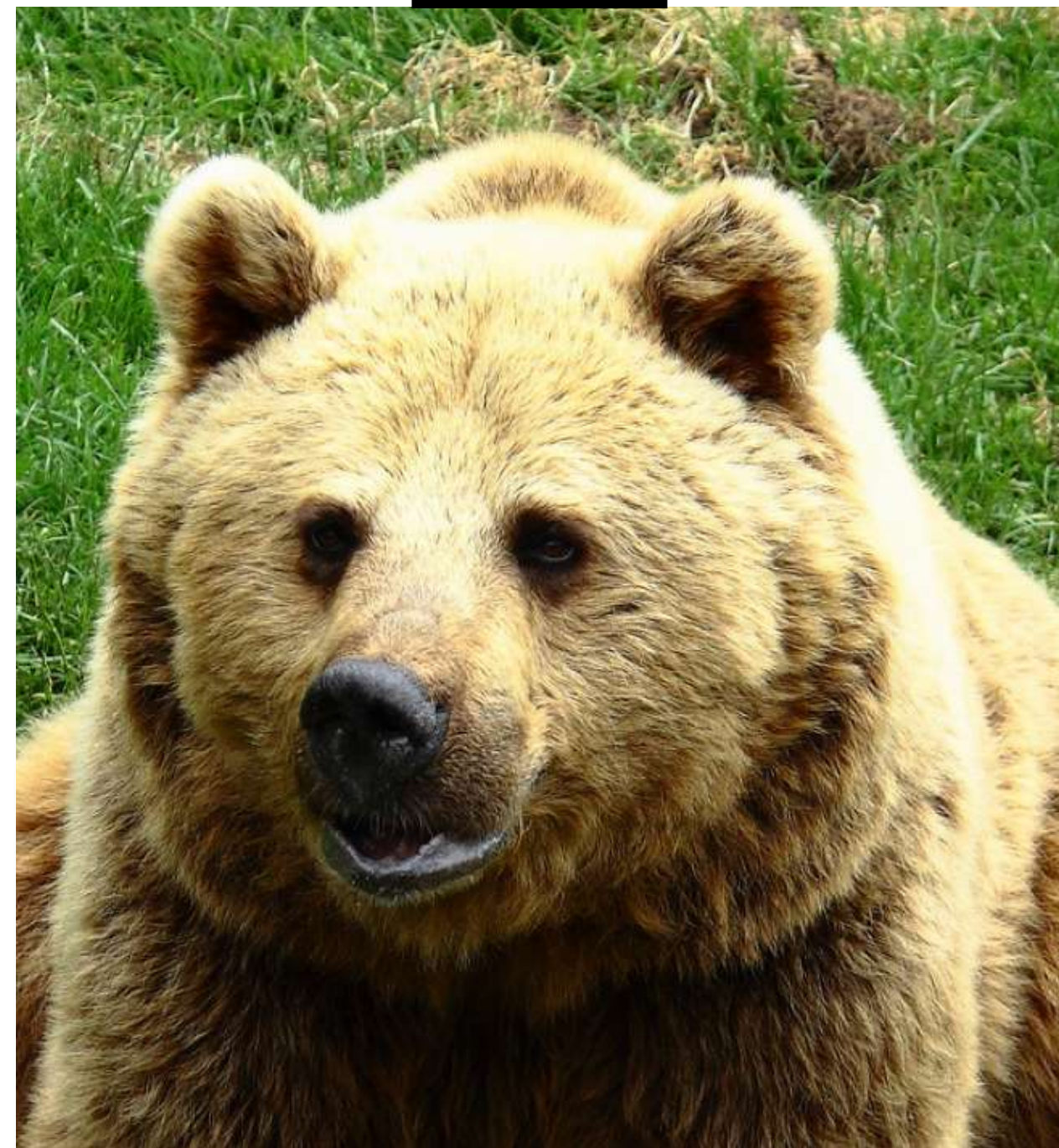


CRATTT (BARON + Oracle TTRV)
Verified: 0 | Rejected: 8 | mAP: 0.0000



Corruption: Gaussian Noise | Severity: 5 | $\tau=0.321$

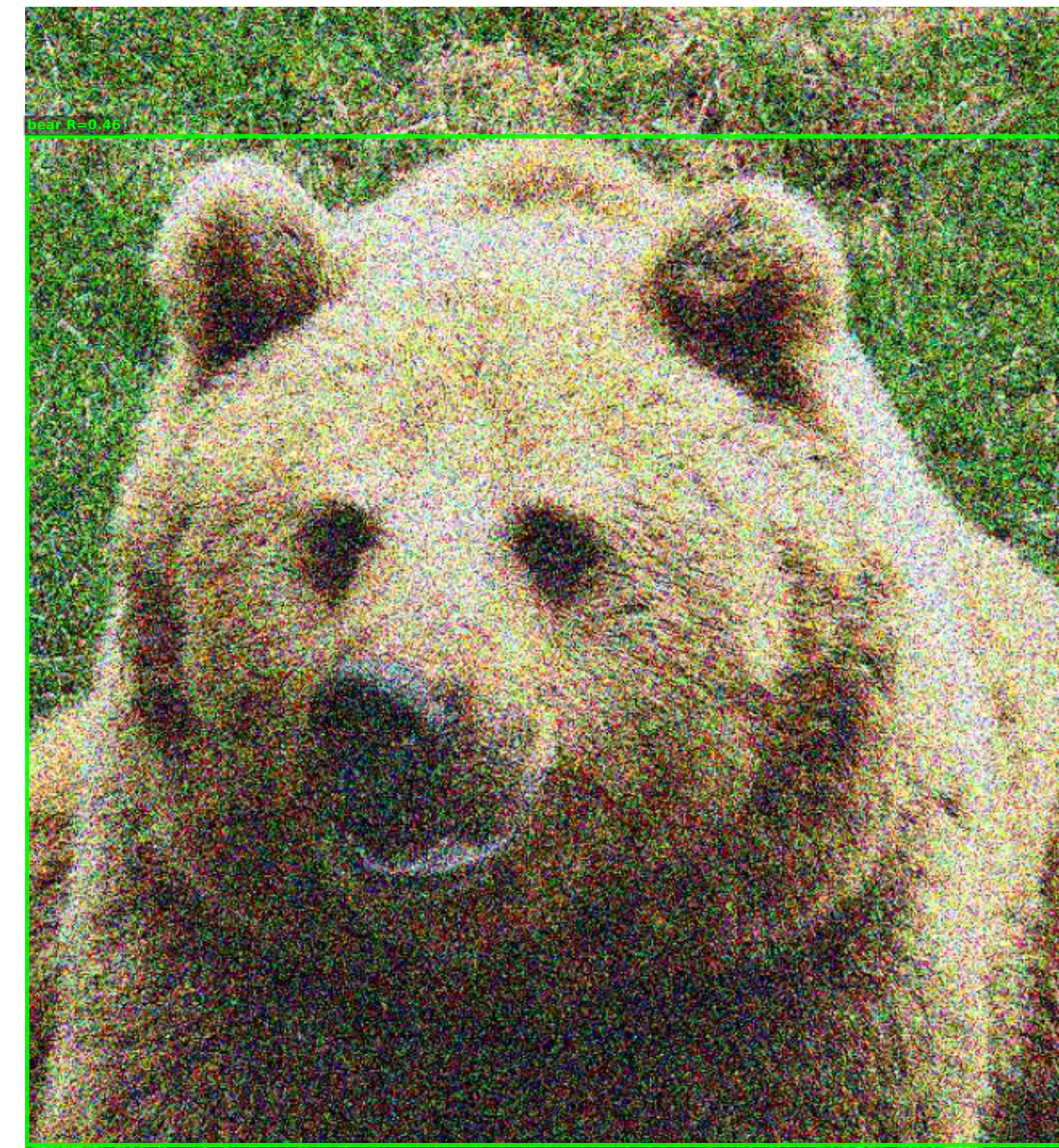
CLEAN REFERENCE
000000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000



Corruption: Gaussian Noise | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 37 | mAP: 0.3447



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 14 | mAP: 0.2250



Corruption: Gaussian Noise | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 0.1683

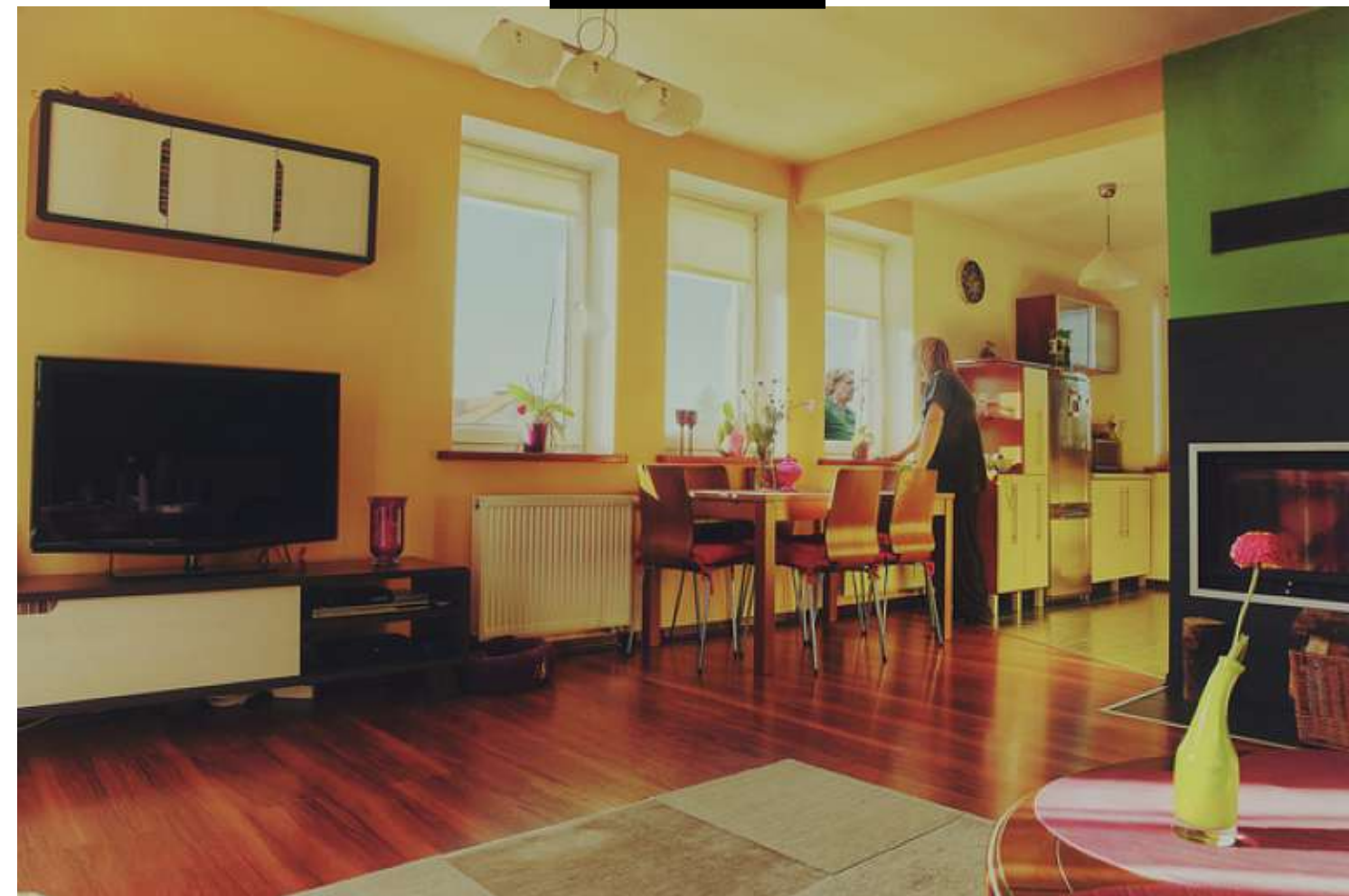


CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 0.1683



Corruption: Motion Blur | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000139.jpg



BASELINE (GroundingDINO)
Detections: 6 | mAP: 0.0123

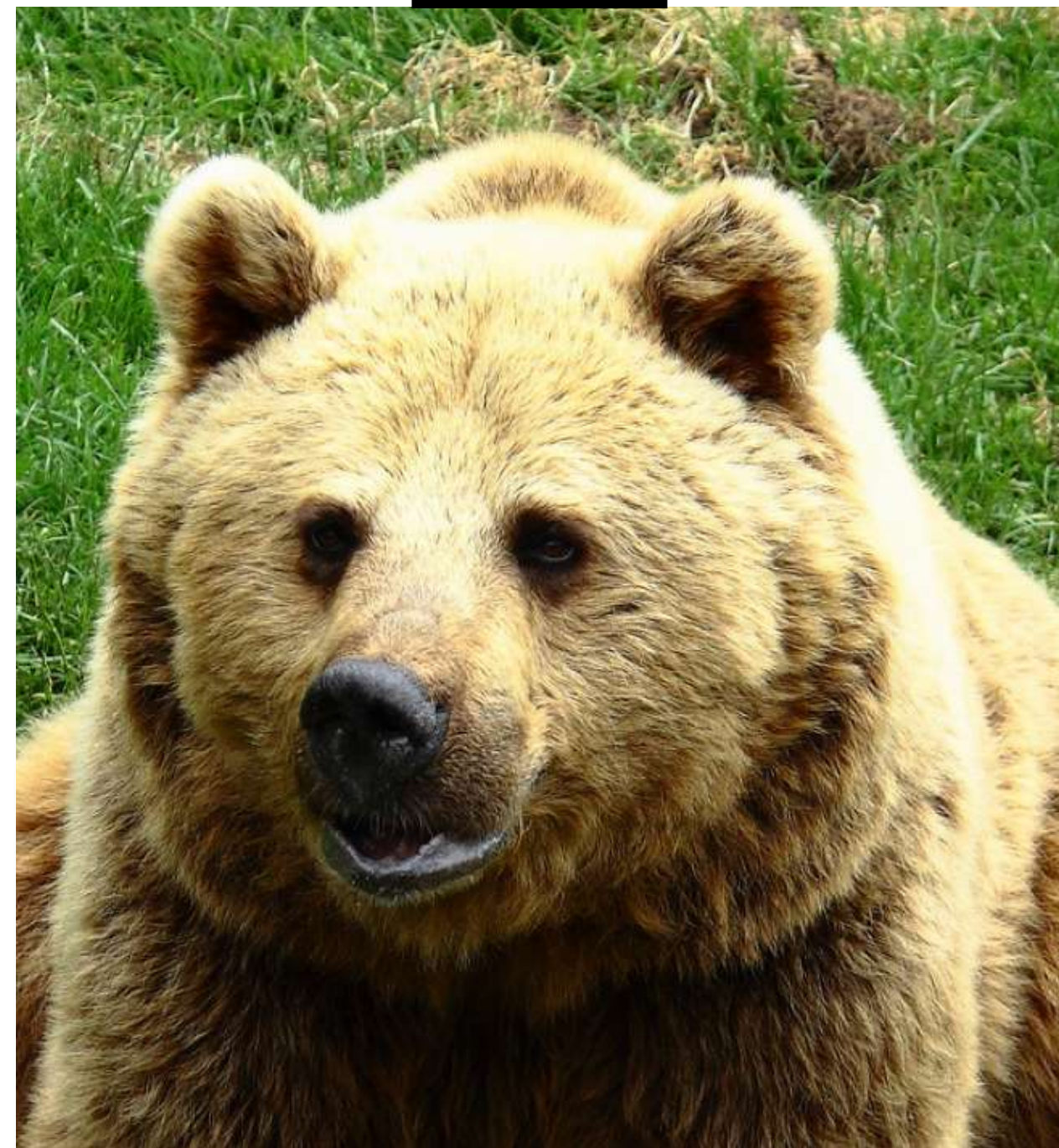


CRATTT (BARON + Oracle TTRV)
Verified: 0 | Rejected: 6 | mAP: 0.0000

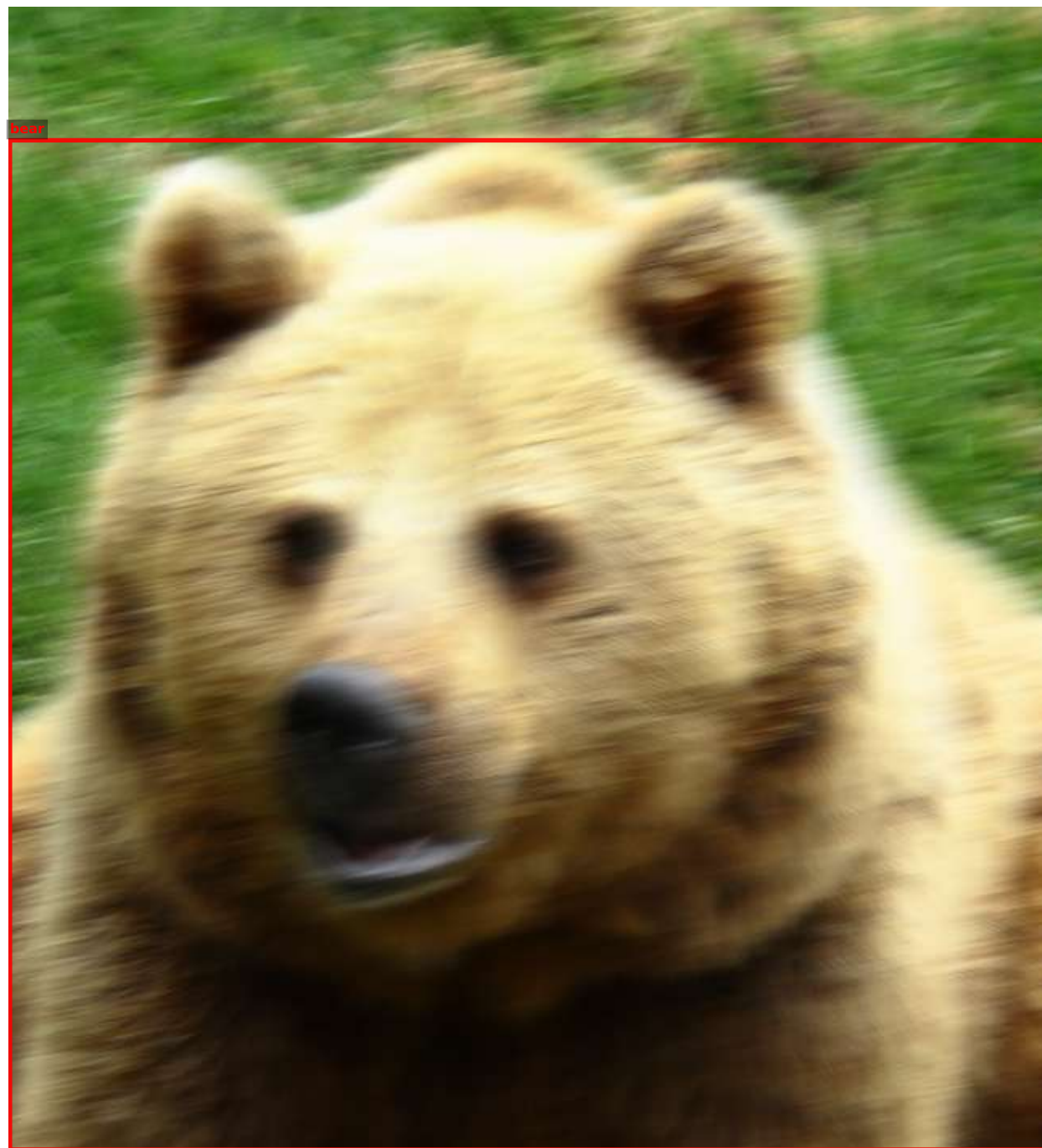


Corruption: Motion Blur | Severity: 5 | $\tau=0.321$

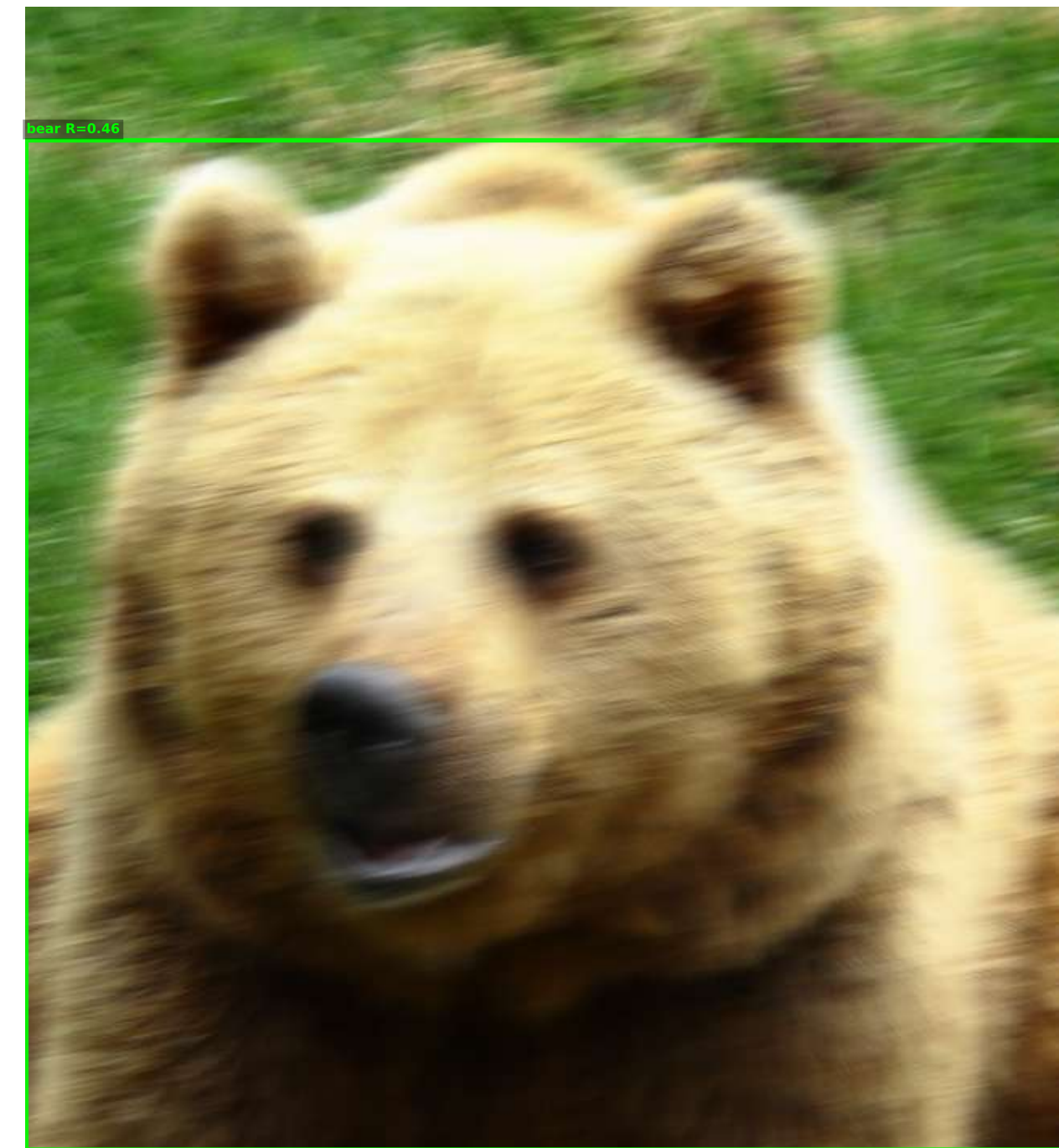
CLEAN REFERENCE
00000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000

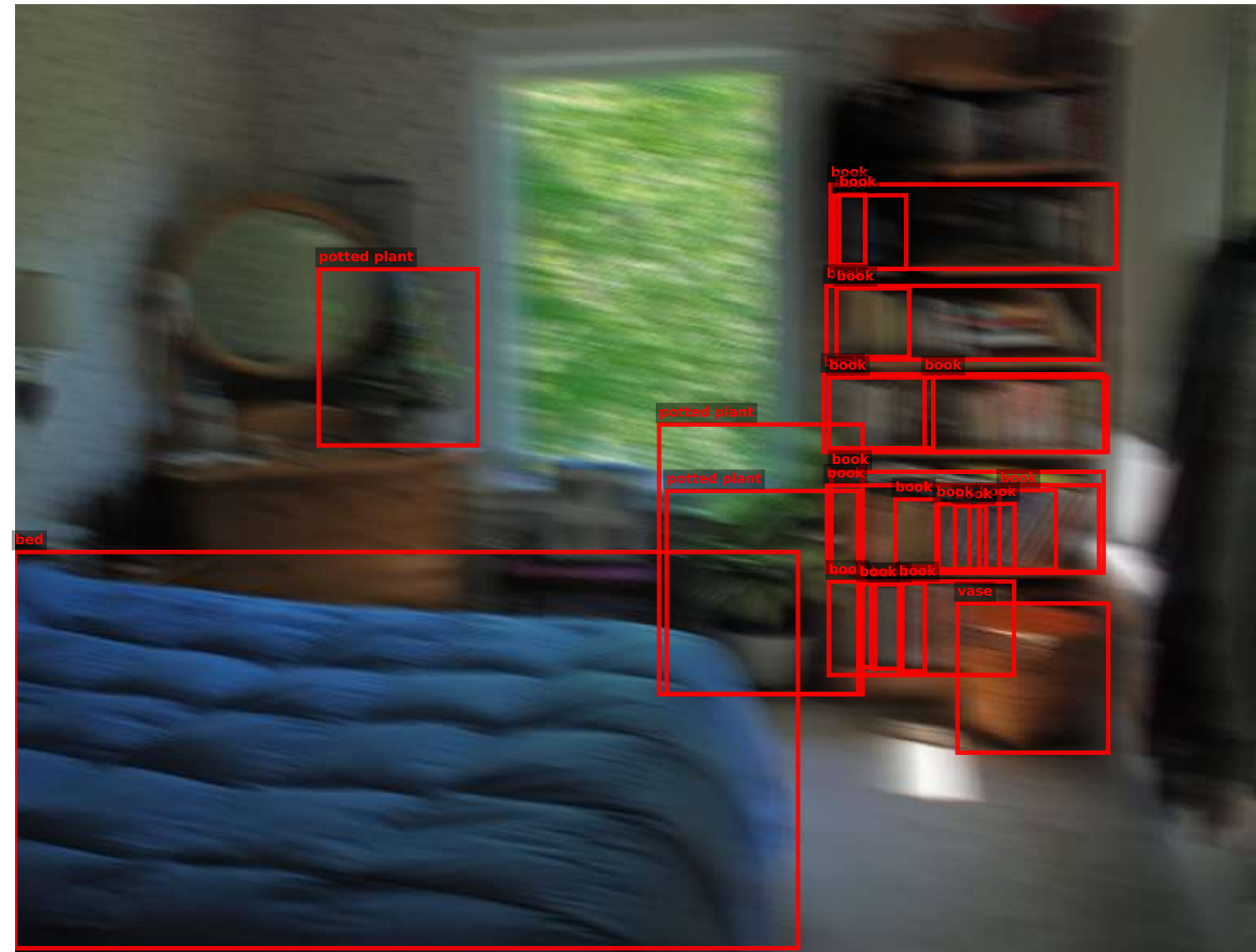


Corruption: Motion Blur | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 24 | mAP: 0.3090



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 14 | mAP: 0.2250



Corruption: Motion Blur | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 0.1347



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 0.1347



Corruption: Fog | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000139.jpg



BASELINE (GroundingDINO)
Detections: 25 | mAP: 0.2998

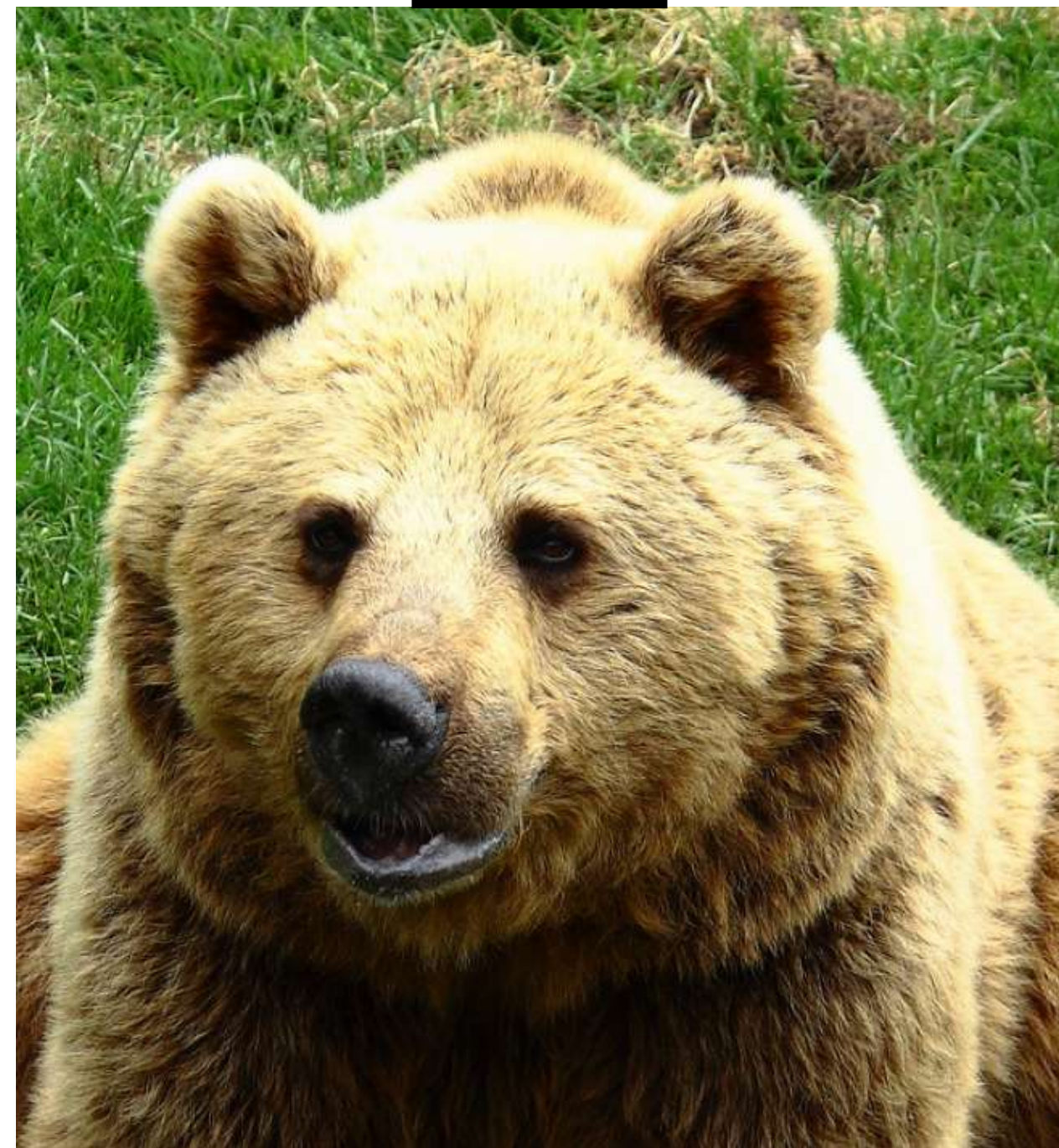


CRATTT (BARON + Oracle TTRV)
Verified: 8 | Rejected: 7 | mAP: 0.2251

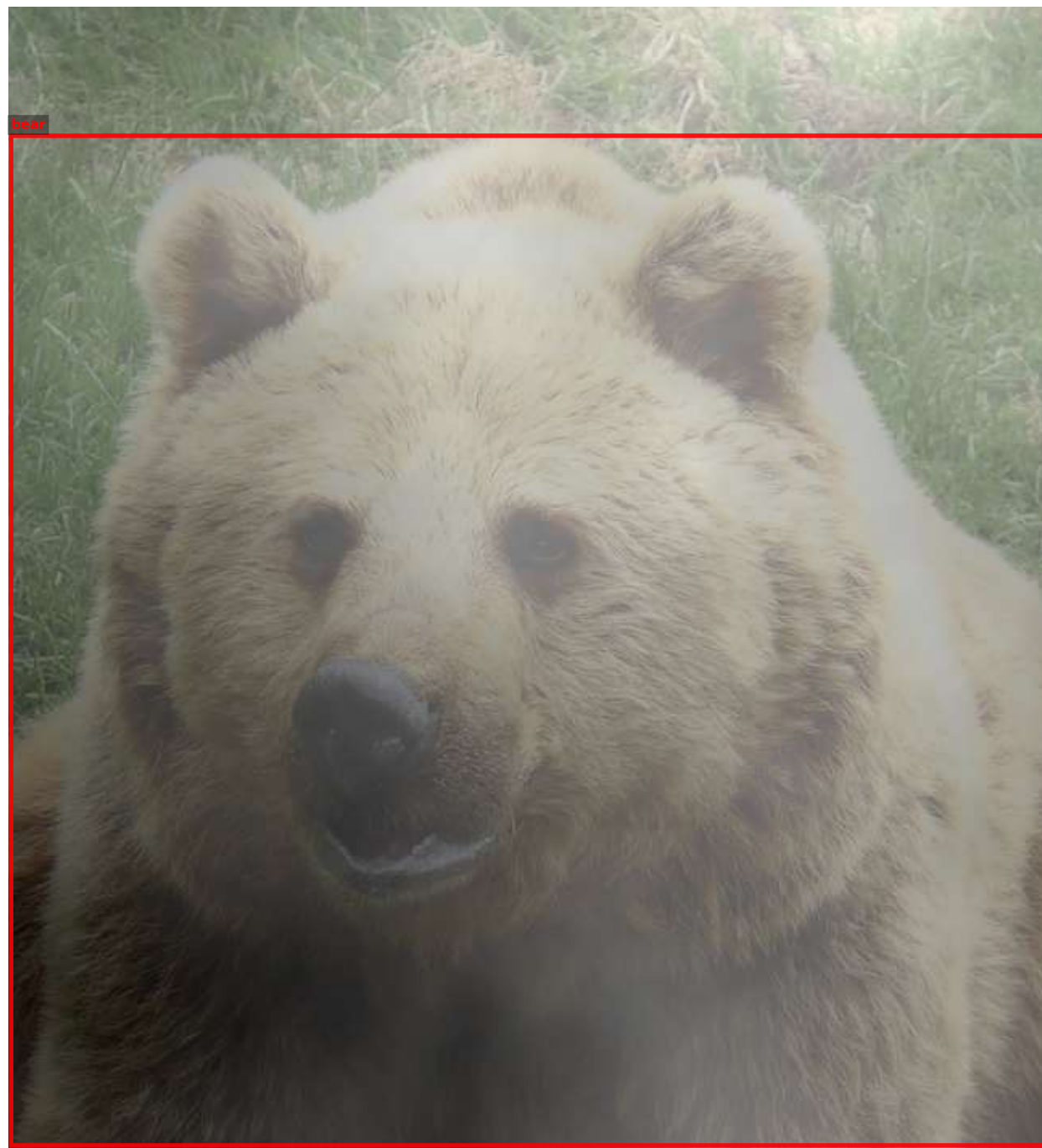


Corruption: Fog | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
00000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000



Corruption: Fog | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 45 | mAP: 0.6860



CRATTT (BARON + Oracle TTRV)
Verified: 4 | Rejected: 11 | mAP: 0.6313



Corruption: Fog | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg

BASELINE (GroundingDINO)
Detections: 4 | mAP: 0.1683

CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 3 | mAP: 0.1683



CLEAN REFERENCE
000000000139.jpg

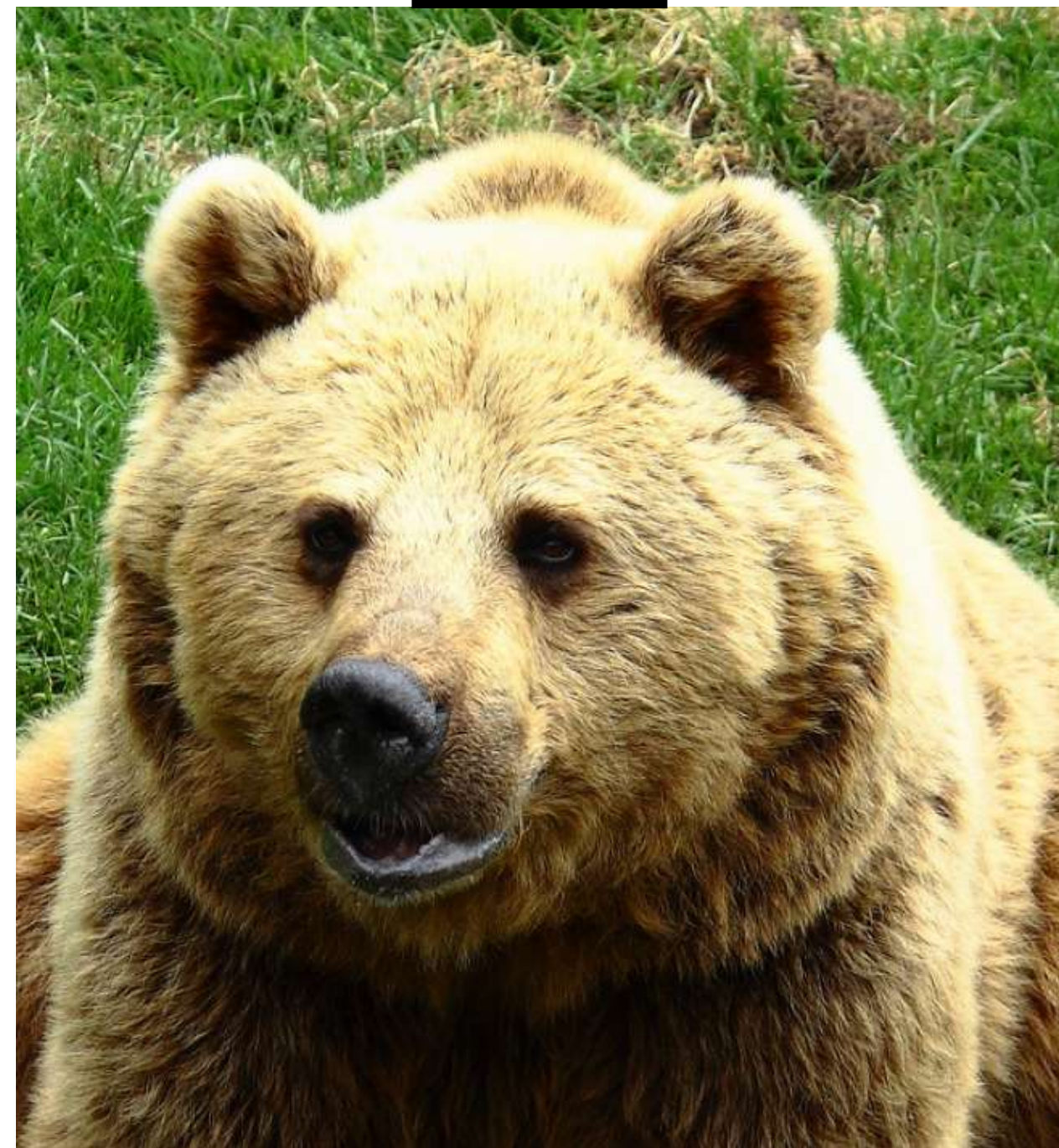
BASELINE (GroundingDINO)
Detections: 21 | mAP: 0.2926

CRATTT (BARON + Oracle TTRV)
Verified: 5 | Rejected: 10 | mAP: 0.1424

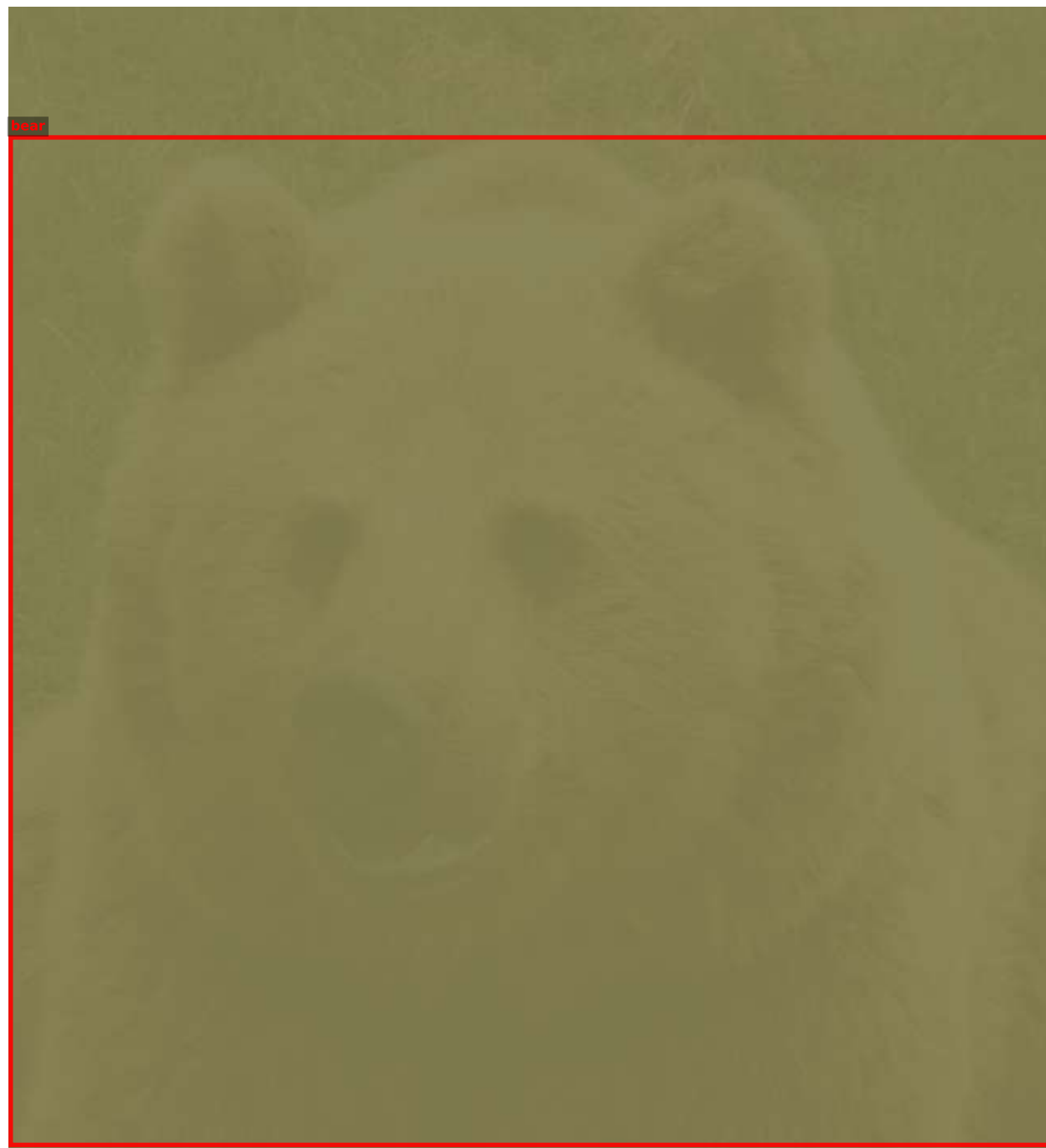


Corruption: Contrast | Severity: 5 | $\tau=0.321$

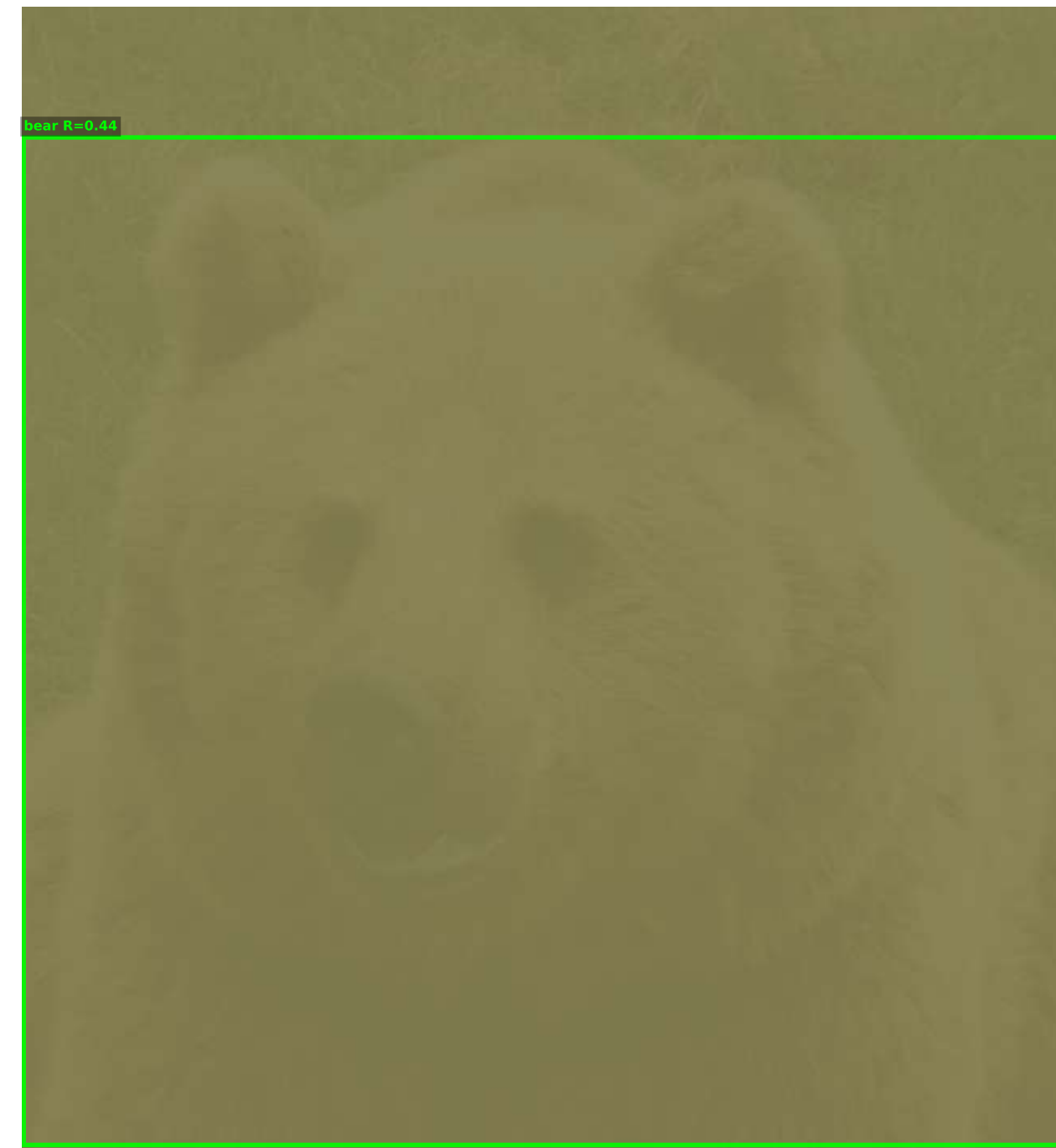
CLEAN REFERENCE
000000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000

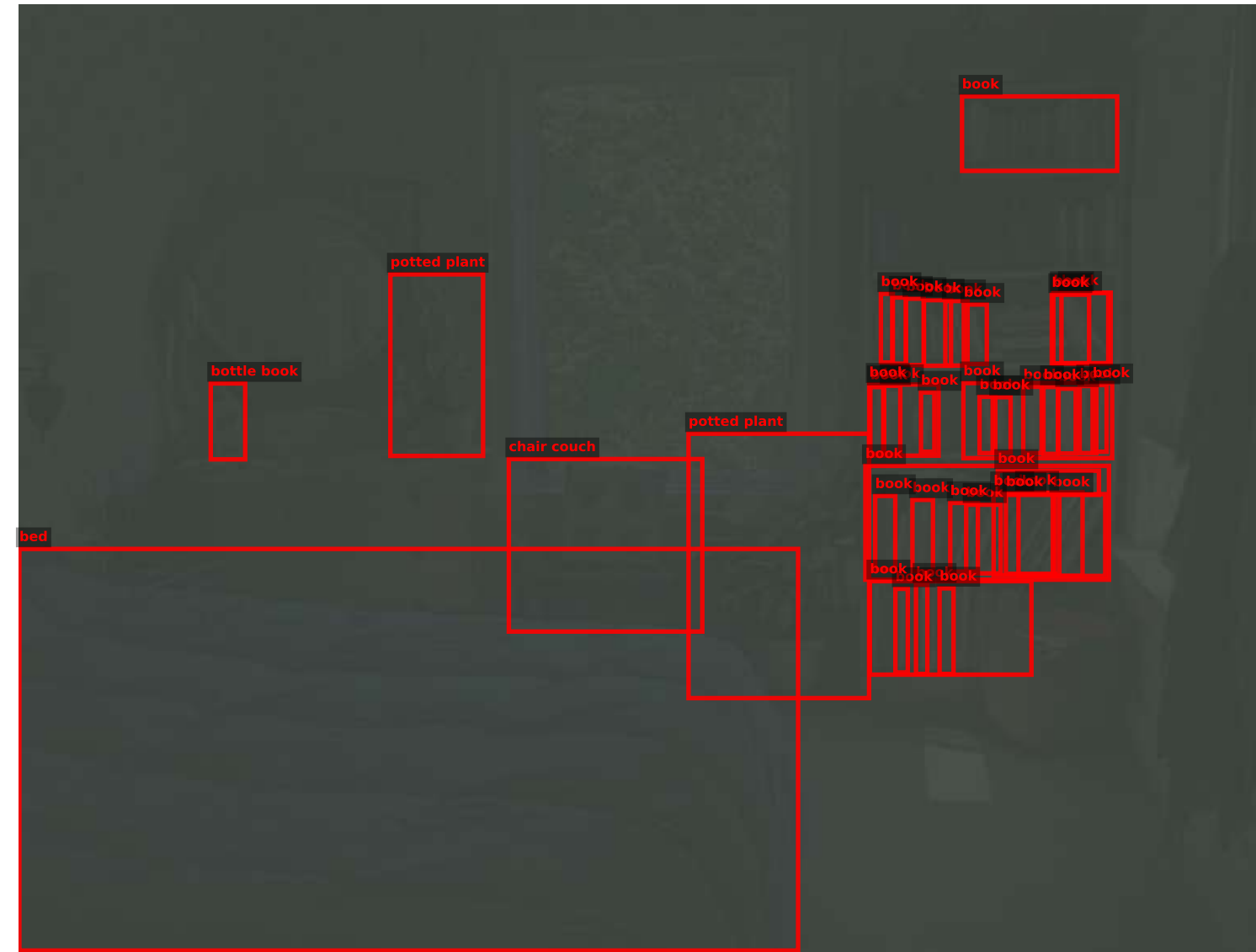


Corruption: Contrast | Severity: 5 | $\tau=0.321$

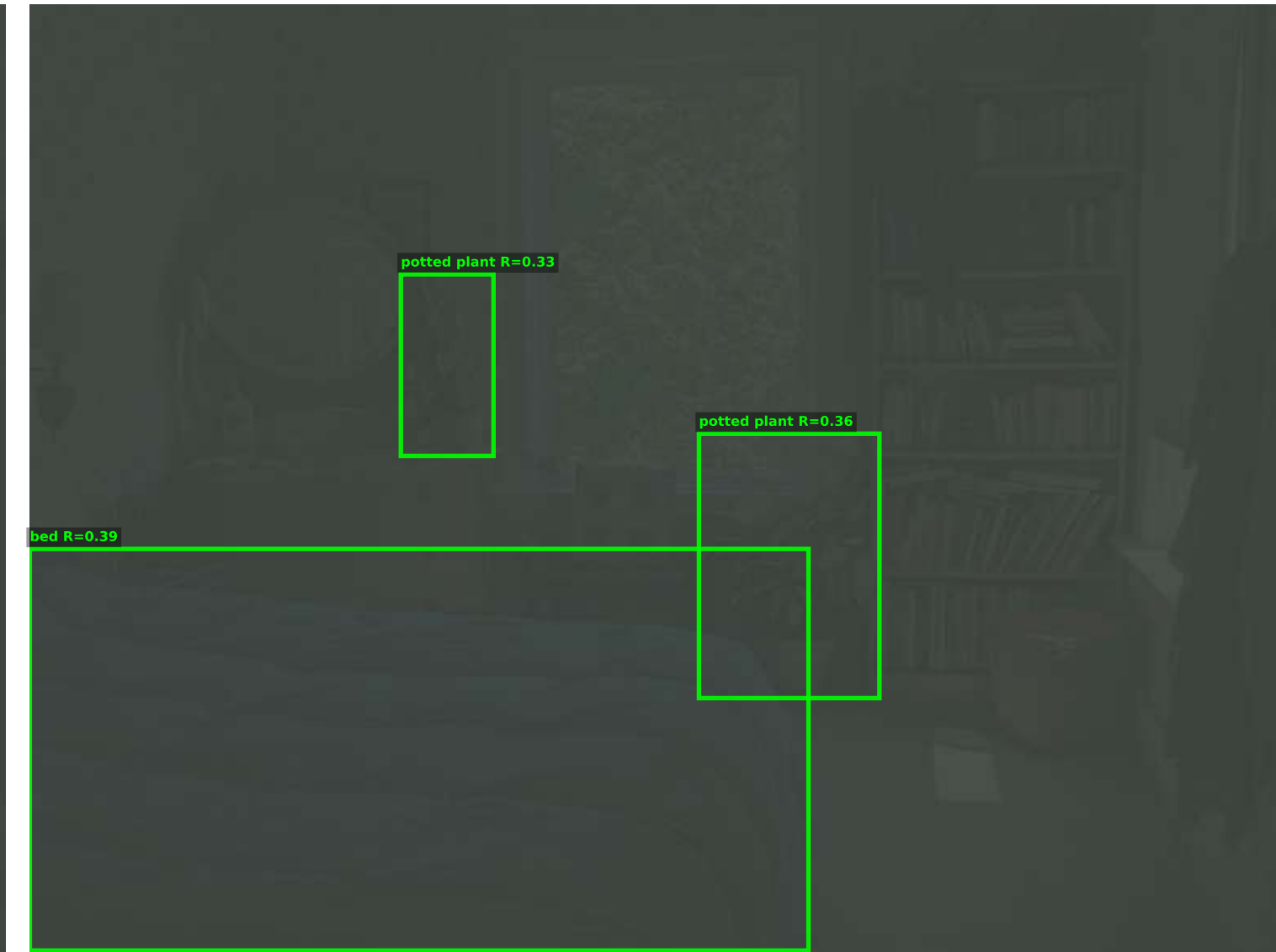
CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 41 | mAP: 0.4700



CRATTT (BARON + Oracle TTRV)
Verified: 3 | Rejected: 12 | mAP: 0.4002

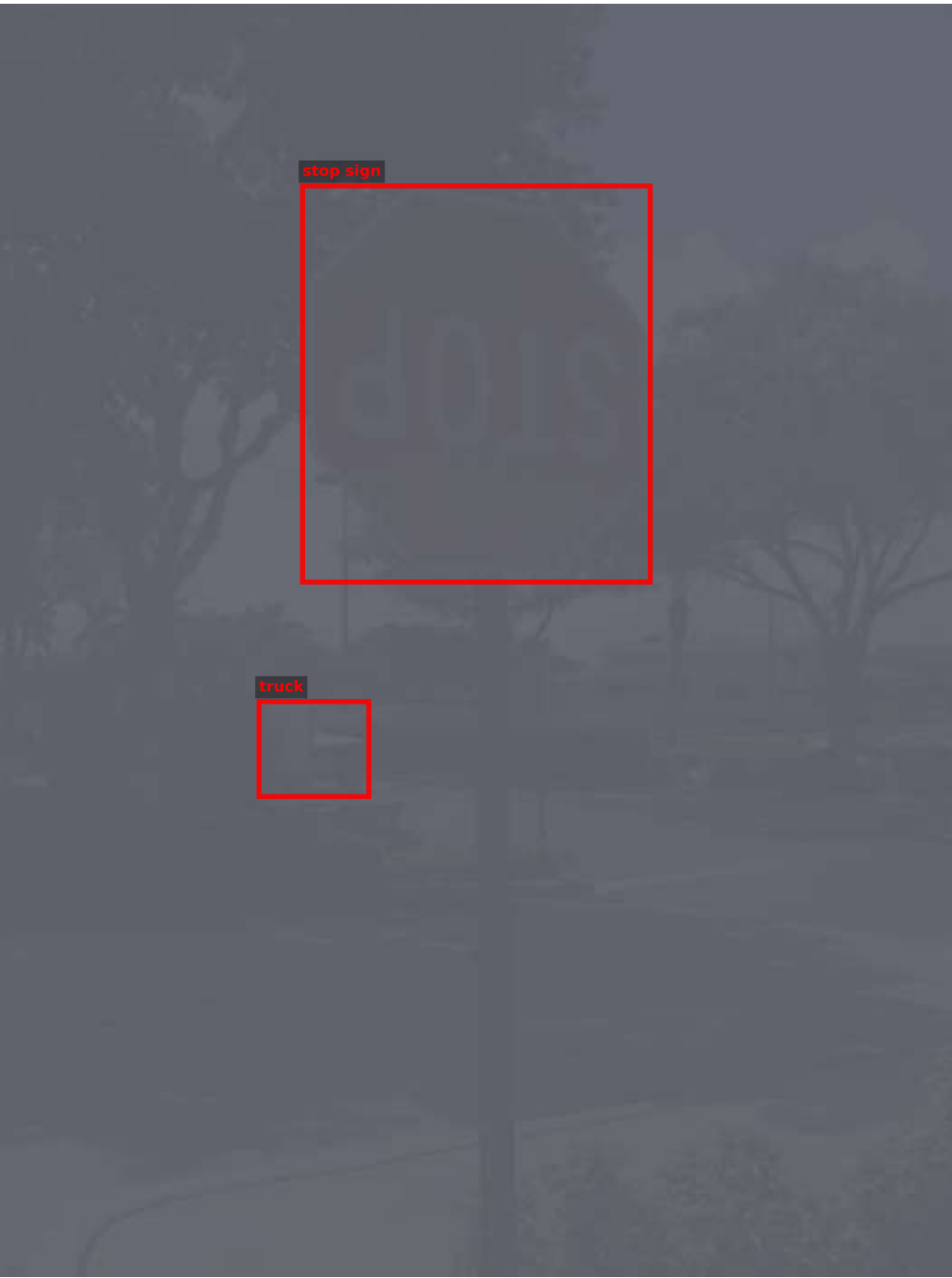


Corruption: Contrast | Severity: 5 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg



BASELINE (GroundingDINO)
Detections: 2 | mAP: 0.1683

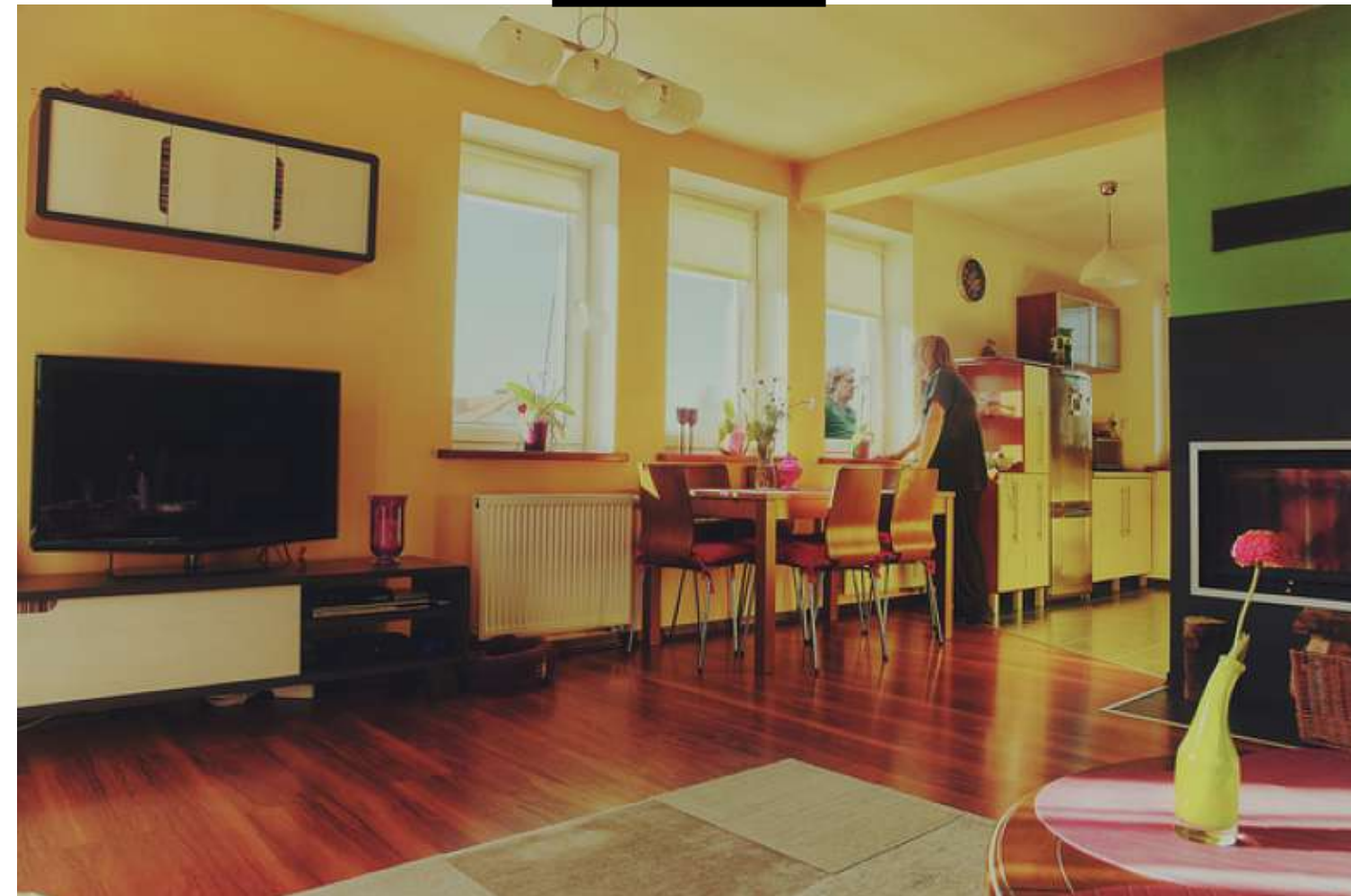


CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 1 | mAP: 0.1683



Corruption: Gaussian Noise | Severity: 3 | $\tau=0.321$

CLEAN REFERENCE
00000000139.jpg



BASELINE (GroundingDINO)
Detections: 19 | mAP: 0.2612

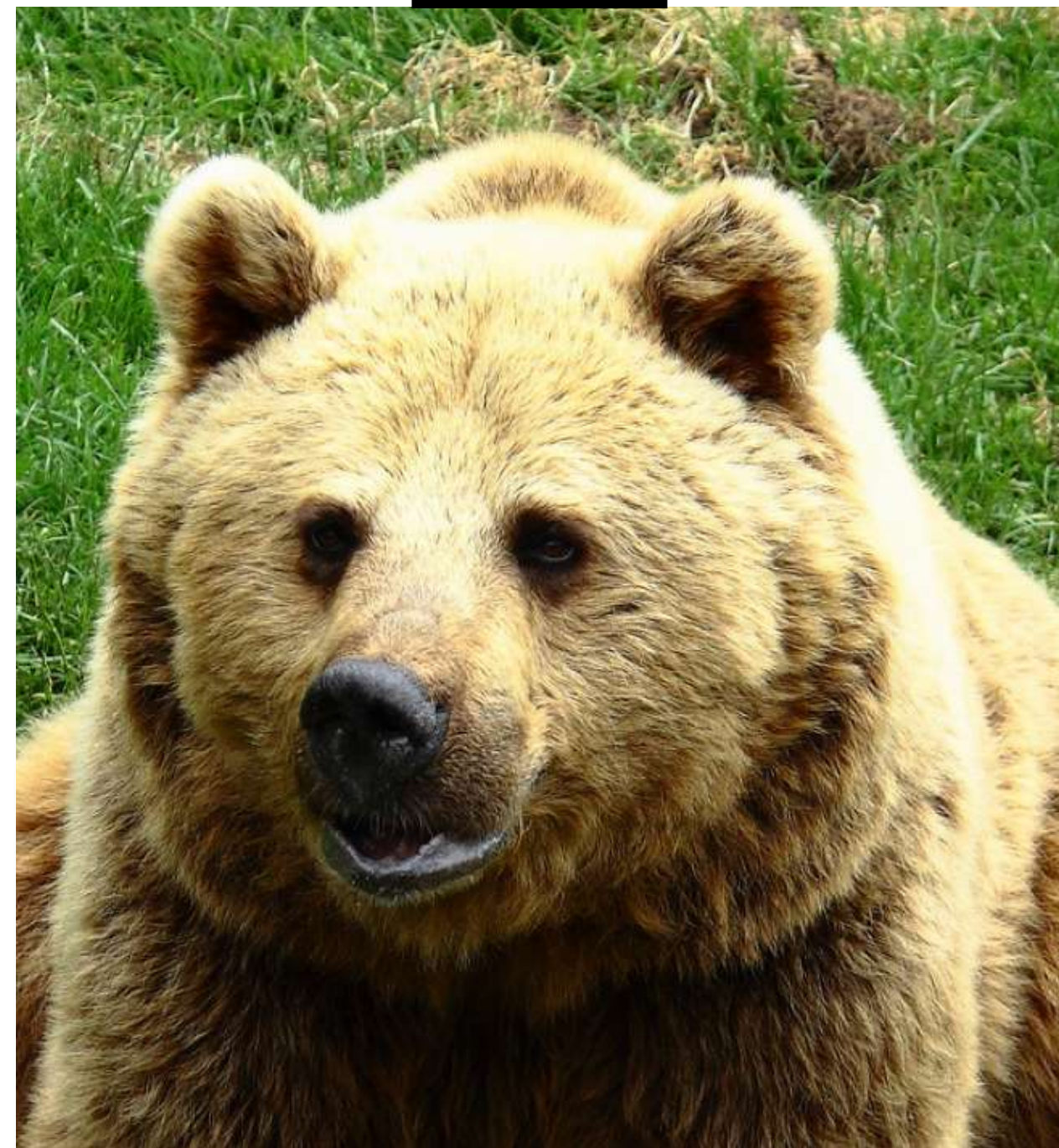


CRATT (BARON + Oracle TTRV)
Verified: 6 | Rejected: 9 | mAP: 0.1847



Corruption: Gaussian Noise | Severity: 3 | $\tau=0.321$

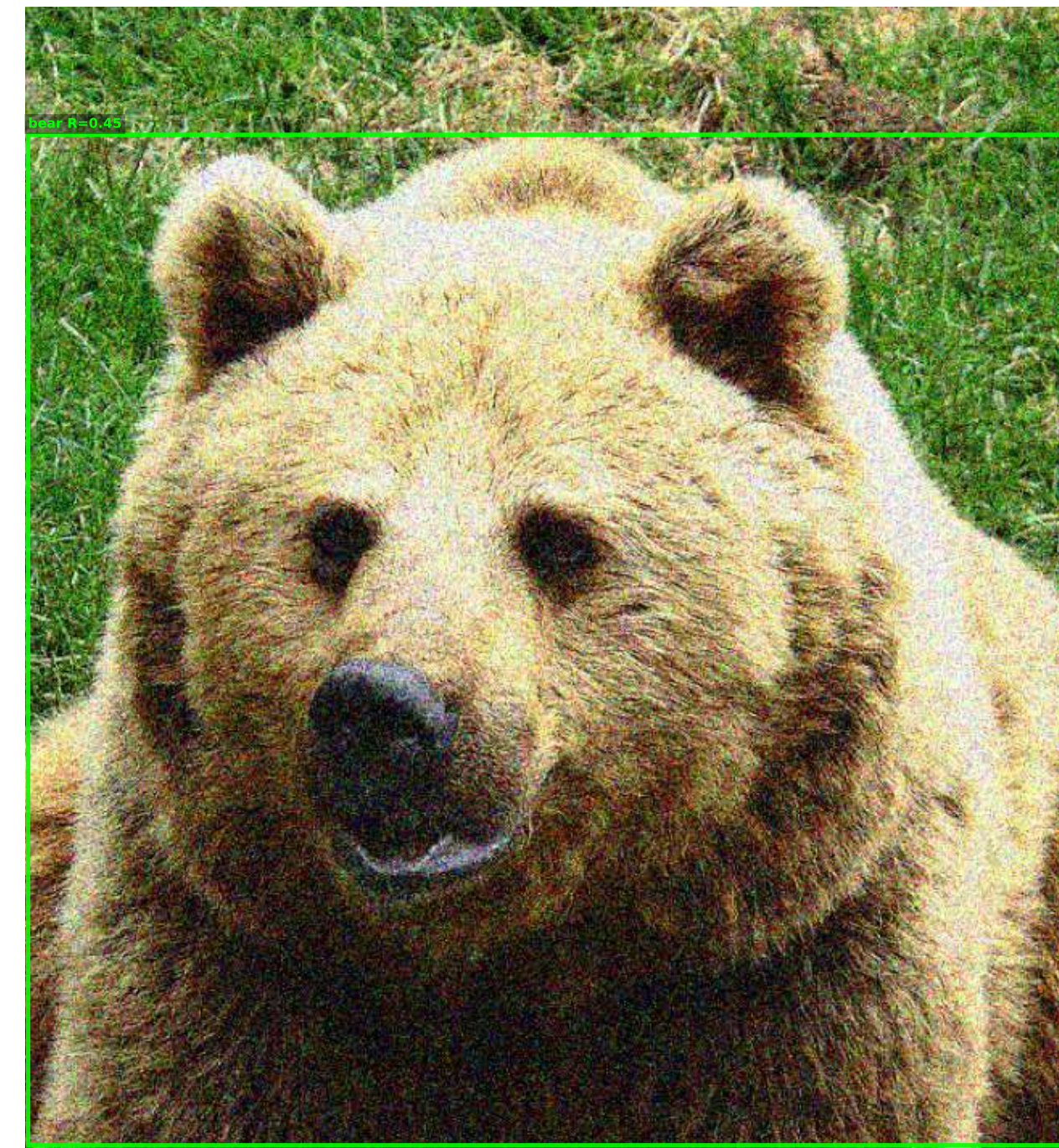
CLEAN REFERENCE
000000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000



Corruption: Gaussian Noise | Severity: 3 | $\tau=0.321$

CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 47 | mAP: 0.4238



CRATTT (BARON + Oracle TTRV)
Verified: 3 | Rejected: 12 | mAP: 0.3505



Corruption: Gaussian Noise | Severity: 3 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg

BASELINE (GroundingDINO)
Detections: 2 | mAP: 0.1683

CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 1 | mAP: 0.1683

stop sign

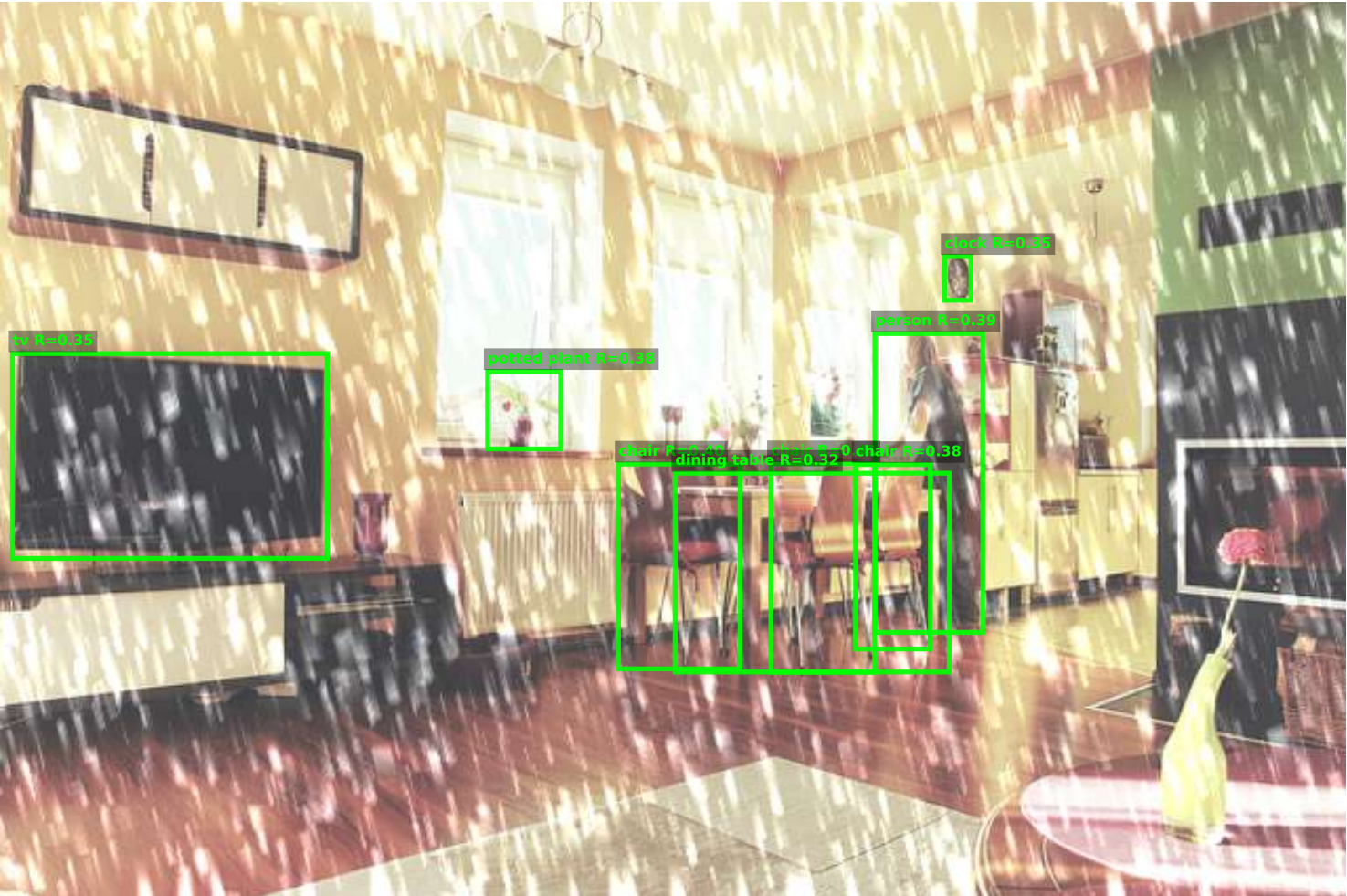
bench

stop sign R=0.45

CLEAN REFERENCE
000000000139.jpg

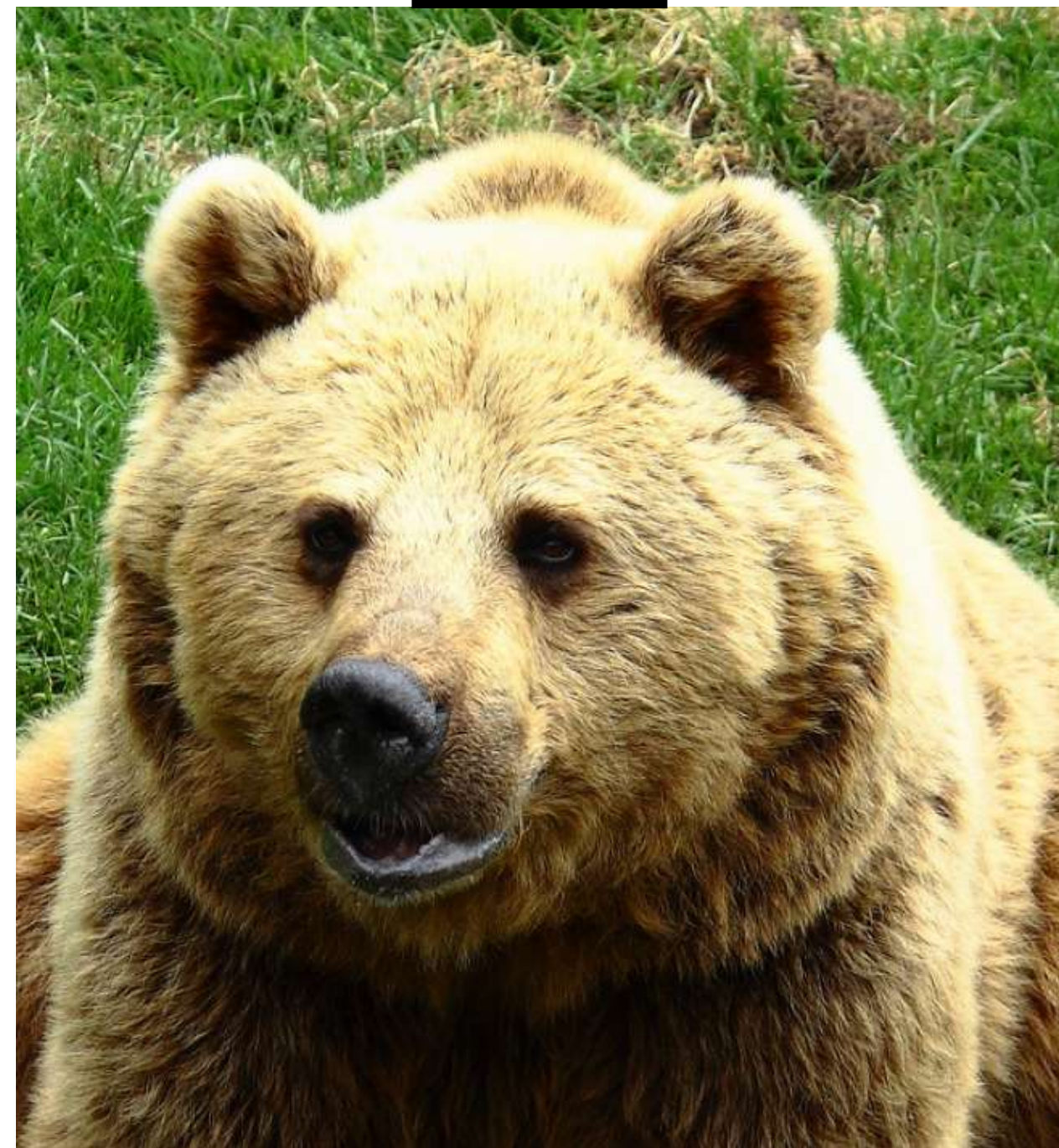
BASELINE (GroundingDINO)
Detections: 24 | mAP: 0.3201

CRATTT (BARON + Oracle TTRV)
Verified: 8 | Rejected: 7 | mAP: 0.2863

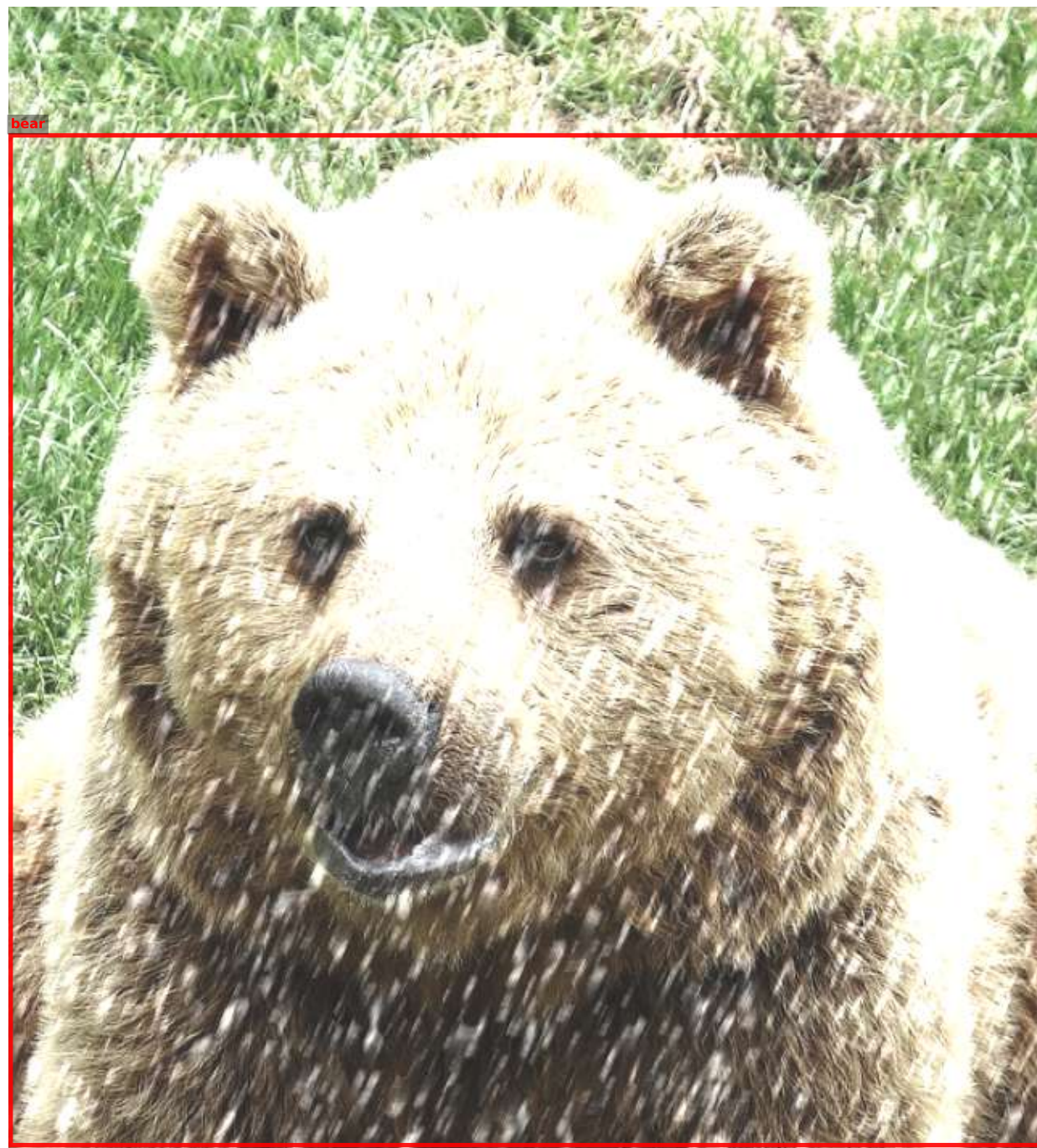


Corruption: Snow | Severity: 3 | $\tau=0.321$

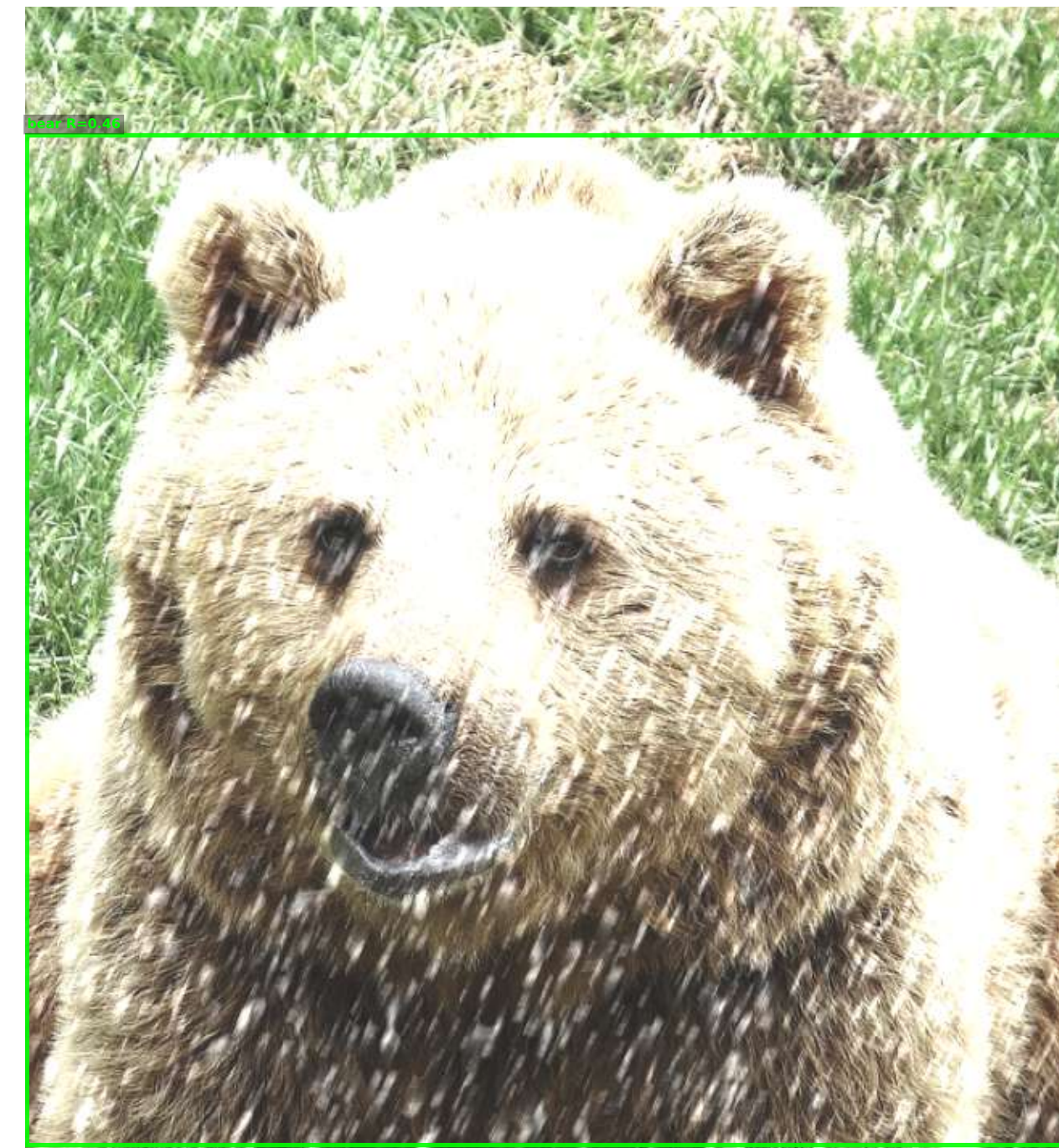
CLEAN REFERENCE
000000000285.jpg



BASELINE (GroundingDINO)
Detections: 1 | mAP: 1.0000



CRATT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 0 | mAP: 1.0000



Corruption: Snow | Severity: 3 | $\tau=0.321$

CLEAN REFERENCE
00000000632.jpg



BASELINE (GroundingDINO)
Detections: 53 | mAP: 0.4479



CRATT (BARON + Oracle TTRV)
Verified: 3 | Rejected: 12 | mAP: 0.3876



Corruption: Snow | Severity: 3 | $\tau=0.321$

CLEAN REFERENCE
000000000724.jpg



BASELINE (GroundingDINO)
Detections: 2 | mAP: 0.1683



CRATTT (BARON + Oracle TTRV)
Verified: 1 | Rejected: 1 | mAP: 0.1683



Gallery Summary: All Visual Comparison Results

Corruption	Sev	Image	Baseline mAP	CRATTT mAP	Δ	Base Dets	CRATTT Dets	Rejected
gaussian_noise	5	000000000139.jpg	0.0744	0.0	-0.0744	8	0	8
gaussian_noise	5	000000000285.jpg	1.0	1.0	0.0	1	1	0
gaussian_noise	5	000000000632.jpg	0.3447	0.225	-0.1197	37	1	14
gaussian_noise	5	000000000724.jpg	0.1683	0.1683	0.0	1	1	0
motion_blur	5	000000000139.jpg	0.0123	0.0	-0.0123	6	0	6
motion_blur	5	000000000285.jpg	1.0	1.0	0.0	1	1	0
motion_blur	5	000000000632.jpg	0.309	0.225	-0.084	24	1	14
motion_blur	5	000000000724.jpg	0.1347	0.1347	0.0	1	1	0
fog	5	000000000139.jpg	0.2998	0.2251	-0.0746	25	8	7
fog	5	000000000285.jpg	1.0	1.0	0.0	1	1	0
fog	5	000000000632.jpg	0.686	0.6313	-0.0547	45	4	11
fog	5	000000000724.jpg	0.1683	0.1683	0.0	4	1	3
contrast	5	000000000139.jpg	0.2926	0.1424	-0.1502	21	5	10
contrast	5	000000000285.jpg	1.0	1.0	0.0	1	1	0
contrast	5	000000000632.jpg	0.47	0.4002	-0.0698	41	3	12
contrast	5	000000000724.jpg	0.1683	0.1683	0.0	2	1	1
gaussian_noise	3	000000000139.jpg	0.2612	0.1847	-0.0766	19	6	9
gaussian_noise	3	000000000285.jpg	1.0	1.0	0.0	1	1	0
gaussian_noise	3	000000000632.jpg	0.4238	0.3505	-0.0733	47	3	12
gaussian_noise	3	000000000724.jpg	0.1683	0.1683	0.0	2	1	1
snow	3	000000000139.jpg	0.3201	0.2863	-0.0338	24	8	7
snow	3	000000000285.jpg	1.0	1.0	0.0	1	1	0
snow	3	000000000632.jpg	0.4479	0.3876	-0.0603	53	3	12
snow	3	000000000724.jpg	0.1683	0.1683	0.0	2	1	1